

**SAFETY DATA SHEET**

Creation Date 16-November-2010

Revision Date 28-March-2024

Revision Number 4

**1. Identification**

**Product Name** 2-Phenoxyethanol

**Cat No. :** A10786

**CAS-No** 122-99-6  
**Synonyms** Ethylene glycol monophenyl ether

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

**Details of the supplier of the safety data sheet****Company****Importer/Distributor**

Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

**Emergency Telephone Number**For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887**2. Hazard(s) identification****Classification****WHMIS 2015 Classification**

Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

|                                                         |            |
|---------------------------------------------------------|------------|
| <b>Acute oral toxicity</b>                              | Category 4 |
| <b>Serious Eye Damage/Eye Irritation</b>                | Category 1 |
| <b>Specific target organ toxicity (single exposure)</b> | Category 3 |
| Target Organs - Respiratory system.                     |            |

**Label Elements****Signal Word**

Danger

**Hazard Statements**

Harmful if swallowed

Causes serious eye damage

May cause respiratory irritation

**Precautionary Statements****Prevention**

Avoid breathing dust/fume/gas/mist/vapors/spray  
Wash face, hands and any exposed skin thoroughly after handling  
Do not eat, drink or smoke when using this product  
Use only outdoors or in a well-ventilated area  
Wear protective gloves/protective clothing/eye protection/face protection

**Response**

IF INHALED: Remove person to fresh air and keep comfortable for breathing  
IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
Immediately call a POISON CENTER/doctor  
Rinse mouth

**Storage**

Store in a well-ventilated place. Keep container tightly closed  
Store locked up

**Disposal**

Dispose of contents/container to an approved waste disposal plant

### 3. Composition/Information on Ingredients

| Component        | CAS-No   | Weight % |
|------------------|----------|----------|
| 2-Phenoxyethanol | 122-99-6 | <= 100   |

### 4. First-aid measures

|                                                               |                                                                                                                   |
|---------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                                         | If symptoms persist, call a physician.                                                                            |
| <b>Eye Contact</b>                                            | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.   |
| <b>Skin Contact</b>                                           | Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician. |
| <b>Inhalation</b>                                             | Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.      |
| <b>Ingestion</b>                                              | Clean mouth with water and drink afterwards plenty of water.                                                      |
| <b>Most important symptoms/effects<br/>Notes to Physician</b> | Causes severe eye damage.<br>Treat symptomatically                                                                |

### 5. Fire-fighting measures

|                                       |                                                                              |
|---------------------------------------|------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>   | Water spray. Carbon dioxide (CO <sub>2</sub> ). Dry chemical. Chemical foam. |
| <b>Unsuitable Extinguishing Media</b> | No information available                                                     |
| <b>Flash Point</b>                    | 130 °C / 266 °F                                                              |

Method - No information available

Autoignition Temperature 430 °C / 806 °F

#### Explosion Limits

Upper No data available

Lower No data available

Sensitivity to Mechanical Impact No information available

Sensitivity to Static Discharge No information available

#### Specific Hazards Arising from the Chemical

Keep product and empty container away from heat and sources of ignition.

#### Hazardous Combustion Products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>).

#### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

#### NFPA

Health  
3

Flammability  
1

Instability  
0

Physical hazards  
N/A

### 6. Accidental release measures

**Personal Precautions** Ensure adequate ventilation. Use personal protective equipment as required.

**Environmental Precautions** Should not be released into the environment.

**Methods for Containment and Clean Up** Keep in suitable, closed containers for disposal. Soak up with inert absorbent material.

### 7. Handling and storage

**Handling** Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation.

**Storage.** Keep in a dry, cool and well-ventilated place. Keep container tightly closed. Incompatible Materials. Strong oxidizing agents. Acid anhydrides. Acid chlorides.

### 8. Exposure controls / personal protection

**Exposure Guidelines** This product does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies.

| Component        | Alberta | British Columbia | Ontario TWAEV                                     | Quebec | ACGIH TLV | OSHA PEL | NIOSH |
|------------------|---------|------------------|---------------------------------------------------|--------|-----------|----------|-------|
| 2-Phenoxyethanol |         |                  | TWA: 25 ppm<br>TWA: 141 mg/m <sup>3</sup><br>Skin |        |           |          |       |

#### Engineering Measures

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

#### Personal protective equipment

**Eye Protection**  
**Hand Protection**Goggles  
Protective gloves

| Glove material                                      | Breakthrough time                 | Glove thickness | Glove comments         |
|-----------------------------------------------------|-----------------------------------|-----------------|------------------------|
| Nitrile rubber<br>Neoprene<br>Natural rubber<br>PVC | See manufacturers recommendations | -               | Splash protection only |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

**Respiratory Protection**

No protective equipment is needed under normal use conditions.

**Environmental exposure controls**

No information available.

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

## 9. Physical and chemical properties

|                                               |                              |
|-----------------------------------------------|------------------------------|
| <b>Physical State</b>                         | Liquid                       |
| <b>Appearance</b>                             | Colorless                    |
| <b>Odor</b>                                   | aromatic                     |
| <b>Odor Threshold</b>                         | No information available     |
| <b>pH</b>                                     | 8-9 10g/100ml water/MeOH     |
| <b>Melting Point/Range</b>                    | 11 - 13 °C / 51.8 - 55.4 °F  |
| <b>Boiling Point/Range</b>                    | 237 °C / 458.6 °F @ 760 mmHg |
| <b>Flash Point</b>                            | 130 °C / 266 °F              |
| <b>Evaporation Rate</b>                       | No information available     |
| <b>Flammability (solid,gas)</b>               | Not applicable               |
| <b>Flammability or explosive limits</b>       |                              |
| Upper                                         | No data available            |
| Lower                                         | No data available            |
| <b>Vapor Pressure</b>                         | 0.04 mbar @ 20 °C            |
| <b>Vapor Density</b>                          | 4.8 (Air = 1.0)              |
| <b>Specific Gravity</b>                       | 1.100                        |
| <b>Solubility</b>                             | 30 g/L @ 20 °C               |
| <b>Partition coefficient; n-octanol/water</b> | No data available            |
| <b>Autoignition Temperature</b>               | 430 °C / 806 °F              |
| <b>Decomposition Temperature</b>              | No information available     |
| <b>Viscosity</b>                              | 29 mPa s at 20 °C            |
| <b>Molecular Formula</b>                      | C8 H10 O2                    |
| <b>Molecular Weight</b>                       | 138.17                       |

## 10. Stability and reactivity

|                            |                                              |
|----------------------------|----------------------------------------------|
| <b>Reactive Hazard</b>     | None known, based on information available   |
| <b>Stability</b>           | Stable under recommended storage conditions. |
| <b>Conditions to Avoid</b> | Incompatible products.                       |

**Incompatible Materials** Strong oxidizing agents, Acid anhydrides, Acid chlorides

**Hazardous Decomposition Products** Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>)

**Hazardous Polymerization** Hazardous polymerization does not occur.

**Hazardous Reactions** None under normal processing.

## 11. Toxicological information

### Acute Toxicity

#### Product Information Component Information

| Component        | LD50 Oral                 | LD50 Dermal               | LC50 Inhalation               |
|------------------|---------------------------|---------------------------|-------------------------------|
| 2-Phenoxyethanol | LD50 = 1850 mg/kg ( Rat ) | LD50 = 5 mL/kg ( Rabbit ) | LC50 > 0.057 mg/L ( Rat ) 8 h |

**Toxicologically Synergistic Products** No information available

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

**Irritation** No information available

**Sensitization** No information available

**Carcinogenicity** The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Component        | CAS-No   | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|------------------|----------|------------|------------|------------|------------|------------|
| 2-Phenoxyethanol | 122-99-6 | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** Respiratory system

**STOT - repeated exposure** None known

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** No information available

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** .

## 12. Ecological information

### Ecotoxicity

Do not empty into drains. .

| Component        | Freshwater Algae                                   | Freshwater Fish                                                                                                          | Microtox                                       | Water Flea                               |
|------------------|----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|------------------------------------------|
| 2-Phenoxyethanol | EC50: > 500 mg/L, 72h<br>(Desmodesmus subspicatus) | LC50: = 366 mg/L, 96h static<br>(Pimephales promelas)<br>LC50: 337 - 352 mg/L, 96h<br>flow-through (Pimephales promelas) | EC50 = 32.4 mg/L 5 min<br>EC50 = 880 mg/L 17 h | EC50: > 500 mg/L, 48h<br>(Daphnia magna) |

|                                      |                                                                         |
|--------------------------------------|-------------------------------------------------------------------------|
| <b>Persistence and Degradability</b> | Persistence is unlikely                                                 |
| <b>Bioaccumulation/ Accumulation</b> | No information available.                                               |
| <b>Mobility</b>                      | . Will likely be mobile in the environment due to its water solubility. |

| Component        | log Pow |
|------------------|---------|
| 2-Phenoxyethanol | 1.2     |

### 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

### 14. Transport information

|                 |               |
|-----------------|---------------|
| <b>DOT</b>      | Not regulated |
| <b>TDG</b>      | Not regulated |
| <b>IATA</b>     | Not regulated |
| <b>IMDG/IMO</b> | Not regulated |

### 15. Regulatory information

#### International Inventories

| Component        | CAS-No   | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS    | ELINCS | NLP |
|------------------|----------|-----|------|------|-----------------------------------------------|-----------|--------|-----|
| 2-Phenoxyethanol | 122-99-6 | X   | -    | X    | ACTIVE                                        | 204-589-7 | -      | -   |

| Component        | CAS-No   | IECSC | KECL     | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|------------------|----------|-------|----------|------|------|------|------|-------|-------|
| 2-Phenoxyethanol | 122-99-6 | X     | KE-28257 | X    | X    | X    | X    | X     | X     |

#### Legend:

X - Listed '-' - Not Listed

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

#### Canada

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

| Component        | Canada - National Pollutant Release Inventory (NPRI) | Canadian Environmental Protection Agency (CEPA) - List of Toxic Substances | Canada's Chemicals Management Plan (CEPA) |
|------------------|------------------------------------------------------|----------------------------------------------------------------------------|-------------------------------------------|
| 2-Phenoxyethanol | Part 4 Substance                                     |                                                                            |                                           |

**Legend** NPRI - National Pollutant Release Inventory

#### Other International Regulations

#### Authorisation/Restrictions according to EU REACH

| Component | REACH (1907/2006) - Annex XIV - Substances Subject to | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous | REACH Regulation (EC 1907/2006) article 59 - Candidate |
|-----------|-------------------------------------------------------|--------------------------------------------------------------------|--------------------------------------------------------|
|-----------|-------------------------------------------------------|--------------------------------------------------------------------|--------------------------------------------------------|

|                  | Authorization | Substances                                                         | List of Substances of Very High Concern (SVHC) |
|------------------|---------------|--------------------------------------------------------------------|------------------------------------------------|
| 2-Phenoxyethanol | -             | Use restricted. See item 75.<br>(see link for restriction details) | -                                              |

## REACH links

<https://echa.europa.eu/substances-restricted-under-reach>

## Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component        | CAS-No   | OECD HPV | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|------------------|----------|----------|------------------------------|---------------------------|--------------------------------------------|
| 2-Phenoxyethanol | 122-99-6 | Listed   | Not applicable               | Not applicable            | Not applicable                             |

| Component        | CAS-No   | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|------------------|----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| 2-Phenoxyethanol | 122-99-6 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

## 16. Other information

## Prepared By

Product Safety Department  
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[www.thermofisher.com](http://www.thermofisher.com)

## Creation Date

16-November-2010

## Revision Date

28-March-2024

## Print Date

28-March-2024

## Revision Summary

New emergency telephone response service provider.

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**